

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS:5

Total Marks 50

Duration :60 Minutes Annexure A

Comparative Analysis

Code of Conduct:

I, K. Mounith (192425090), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Mounith

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	

Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
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Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Depth Estimation Comparison (Stereo vs SfM) Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A 2.0 2.5
B 3.0 3.8
C 4.0 5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

A. Depth Estimation Comparison — Stereo vs SfM

Given Data

Object	Actual Depth	Stereo Vision	Structure from Motion (SfM)
A	2.0 m	2.0 m	2.5 m
B	3.0 m	3.0 m	3.8 m
C	4.0 m	4.0 m	5.0 m

(a) Absolute Error

Formula:

Absolute Error=|Estimated Depth–Actual Depth|

Object A

Stereo:

$$|2.0-2.0|=0$$

SfM:

$$|2.5-2.0|=0.5$$

Object B

Stereo:

$$|3.0-3.0|=0$$

SfM:

$$|3.8-3.0|=0.8$$

Object C

Stereo:

$$|4.0-4.0|=0$$

SfM:

$$|5.0-4.0|=1.0$$

Total Error

Stereo:

$$0+0+0=0$$

SfM:

$$0.5+0.8+1.0=2.3$$

(b) Comparison

Method	Total Absolute Error
Stereo Vision	0 m
SfM	2.3 m

Conclusion: Stereo Vision is more accurate because its total absolute error is 0 m, while SfM has an error of 2.3 m.

2. Pattern Recognition Accuracy

Comparison Two recognition systems

produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

A. Pattern Recognition Accuracy Comparison

Given Data

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Accuracy

Formula:

$$\text{Accuracy} = \frac{\text{Total Samples}}{\text{Correct Predictions}} \times 100$$

Traditional System

$$\text{Accuracy} = \frac{100}{85} \times 100 = 85\%$$

Deep Learning

$$\text{Accuracy} = \frac{100}{92} \times 100 = 92\%$$

(b) Comparison

System	Accuracy
Traditional	85%
Deep Learning	92%

Difference:

$$92\% - 85\% = 7\%$$

Conclusion: Deep Learning performs better because it achieves 92% accuracy, which is 7 percentage points higher than the Traditional system.

3. Precision-Recall Trade-off Analysis

Two models produce:

Model	TP	FP	FN
A	70	30	20
B	85	25	25

(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced.

A. Precision-Recall Trade-off Analysis

Given Data

Model	TP	FP	FN
A	70	30	20
B	85	25	25

(a) Precision and Recall

Formulas

Precision = $\frac{TP}{TP+FP}$ Recall = $\frac{TP}{TP+FN}$

Model A

Precision

PrecisionA = $\frac{70}{70+30} = \frac{70}{100} = 0.70 = 70\%$

Recall

RecallA = $\frac{70}{70+20} = \frac{70}{90} = 0.778 = 77.8\%$

Model B

Precision

PrecisionB = $\frac{85}{85+25} = \frac{85}{110} = 0.773 = 77.3\%$

Recall

RecallB = $\frac{85}{85+25} = \frac{85}{110} = 0.773 = 77.3\%$

(b) Comparison

Model	Precision	Recall
A	70%	77.8%
B	77.3%	77.3%

Conclusion: Model B is more balanced because its precision and recall are both 77.3%. Model A has a larger difference between precision (70%) and recall (77.8%).

Therefore, Model B is the better balanced model.

4. F1-Score Comparison Two

models:

Model	Precision	Recall
A	0.8	0.6
B	0.75	0.75

- (a) Compute F1-score for both models.
 (b) Compare and evaluate which model is better.

A. F1-Score Comparison

Given Data

Model	Precision	Recall
A	0.80	0.60
B	0.75	0.75

(a) F1-Score

Formula:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Model A

$$F1A = \frac{2 \times 0.8 \times 0.6}{0.8 + 0.6} = \frac{2 \times 1.40.48}{1.40.96} = 0.686$$

So,

$$F1A \approx 68.6\%$$

Model B

$$F1B = \frac{2 \times 0.75 \times 0.75}{0.75 + 0.75} = \frac{2 \times 1.50.5625}{1.51.125} = 0.75$$

So,

$$F1B = 75\%$$

(b) Comparison

Model	Precision	Recall	F1-Score
A	80%	60%	68.6%
B	75%	75%	75%

Conclusion: Model B is better because its F1-score is 75%, compared with 68.6% for Model A. Model B also has equal precision and recall, showing a better balance.

5. Optical Flow Consistency Motion

values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

A. Optical Flow Consistency

Given Data

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Average Motion

Method A

Formula:

Average = $\frac{\text{Number of frames} \times \text{Sum of motion values}}{\text{Number of frames}}$ Average A = $\frac{4+5+6}{3} = 5$

Method B

Average B = $\frac{6+8+9}{3} = 7.67$

(b) Compare Consistency

Method A values:

4, 5, 6

Changes:

$5-4=1$ $6-5=1$

Method B values:

6, 8, 9

Changes:

$8-6=2$ $9-8=1$

Method A changes gradually by 1 unit in every frame, whereas Method B changes by 2 units and then 1 unit.

Comparison

Method	Average Motion	Consistency
A	5.00	More stable
B	7.67	Less stable

Conclusion: Method A is more stable because its motion values increase consistently by 1 unit per frame.

Method B has a larger change between frames 1 and 2.